

PROJECT REPORT

TITTLE: Emotion & Sentiment Analyzer

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Date: MARCH 2026

1. Introduction

The Emotion & Sentiment Analyzer is a real-time system designed to understand human emotions using both facial expressions and voice input. The idea behind this project was to create something that feels interactive and practical rather than just theoretical.

In everyday life, humans naturally understand emotions by observing expressions and listening to tone. However, computers lack this ability. This project attempts to bridge that gap by combining computer vision and natural language processing into a single application.

The system uses a webcam to capture facial expressions and analyzes them using image processing techniques. At the same time, it records voice input and evaluates the sentiment of spoken words. The final result is generated by combining both inputs, giving a

more complete understanding of the user's emotional state.

The goal was not to build a perfect AI system but to create a working prototype that demonstrates how multimodal systems can improve human-computer interaction.

2. Problem Statement

Most emotion detection systems available today rely on a single type of input, such as only images or only text. This creates limitations because human emotions are complex and cannot always be captured through just one medium.

For example, a person may appear neutral in facial expression but may sound stressed while speaking. In such cases, single-input systems fail to provide accurate results.

The lack of multimodal systems leads to:

- incomplete emotional understanding
- reduced accuracy
- poor real-world applicability
- limited interaction capabilities

Therefore, there is a need for a system that can combine multiple inputs and provide a more reliable emotional analysis.

This project aims to solve that problem by building a system that integrates facial emotion detection and voice sentiment analysis into a unified platform.

3. Functional Requirements

The system includes the following key functionalities:

- Real-time face detection using webcam
- Emotion estimation based on facial features
- Audio recording and speech recognition
- Sentiment analysis of spoken text
- Gesture-based interaction (capture, record, trigger actions)
- Final emotion decision based on combined inputs
- Display of results in a graphical interface
- History tracking of recent emotions

4. Non-Functional Requirements

- The system should be simple and easy to use
- The interface should be clean and responsive
- The system should handle real-time inputs smoothly

- Output should remain stable and not fluctuate rapidly
- Code should be modular for future improvements
- The application should not crash under normal usage

5. System Architecture

The system follows a modular structure where each component handles a specific task:

- Video Module → captures webcam input
- Face Detection Module → detects faces using OpenCV
- Emotion Module → analyzes facial features
- Audio Module → records and processes speech
- Sentiment Module → evaluates text polarity

- Gesture Module → detects hand movements
- Fusion Module → combines results
- UI Module → displays everything to the user

This modular approach makes the system easier to manage and extend in the future.

6. Key Design Decisions

Several important decisions were made during development:

Python was chosen as the primary language due to its simplicity and availability of libraries. Tkinter was used for GUI to keep the system lightweight and easy to run on any machine.

Instead of using heavy deep learning models, a heuristic-based approach was used for

facial emotion detection. This decision was taken to avoid dependency issues and ensure smooth execution on local systems.

Gesture control was implemented using basic computer vision techniques rather than advanced models. The aim was to demonstrate functionality without increasing system complexity.

The freeze mechanism was added to solve the problem of fluctuating outputs, which is common in real-time systems.

7. Implementation Overview

The working of the system can be understood in the following steps:

1. The webcam continuously captures video frames

2. Faces are detected in each frame
3. Facial regions are analyzed to estimate emotion
4. Gestures are detected to trigger actions
5. Audio is recorded when required
6. Speech is converted into text
7. Sentiment is extracted from the text
8. Both results are combined
9. Final output is displayed on the interface

The system uses threading to ensure that audio recording does not block the main application. A queue is used for safe communication between processes.

When an image is uploaded, it is directly processed by the system in the background without requiring any additional steps from the user. The result is then reflected in the output and stored in the recent history for reference.

8. Challenges Faced

During development, several practical challenges were encountered:

- Gesture detection was unstable due to lighting variations
- Real-time processing caused rapid fluctuations in output
- Managing multiple inputs (video + audio) required synchronization
- Avoiding random UI triggers (like file dialogs opening repeatedly)
- Ensuring the system does not freeze or crash

These challenges were addressed by adding smoothing logic, cooldown timers, and a stable update mechanism.

9. Testing

Testing was performed at different levels:

- Individual module testing (face, audio, gesture)
- End-to-end testing (full system execution)
- Edge case testing (no face, no audio, poor lighting)
- User interaction testing

The system was tested multiple times to ensure stable performance during demonstrations.

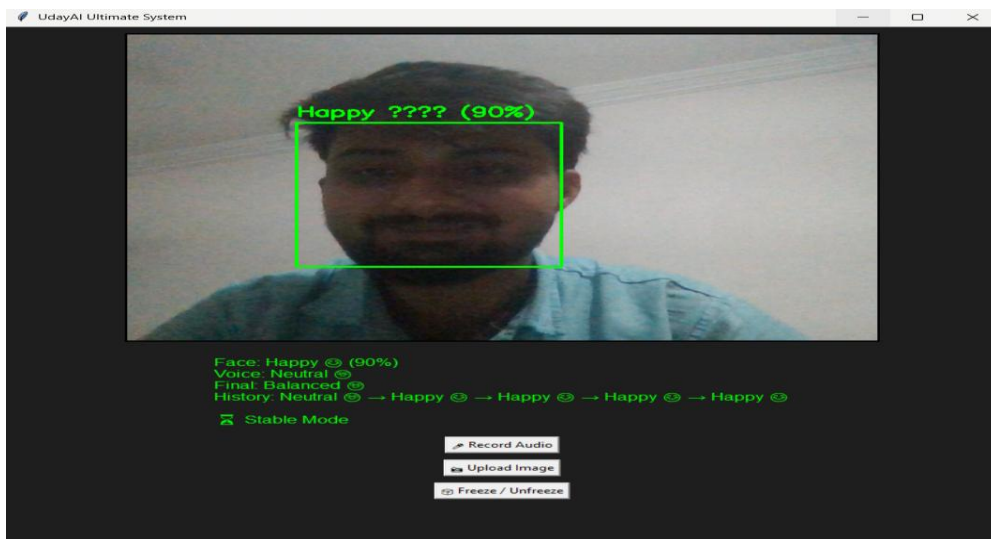
10. Results and Discussion

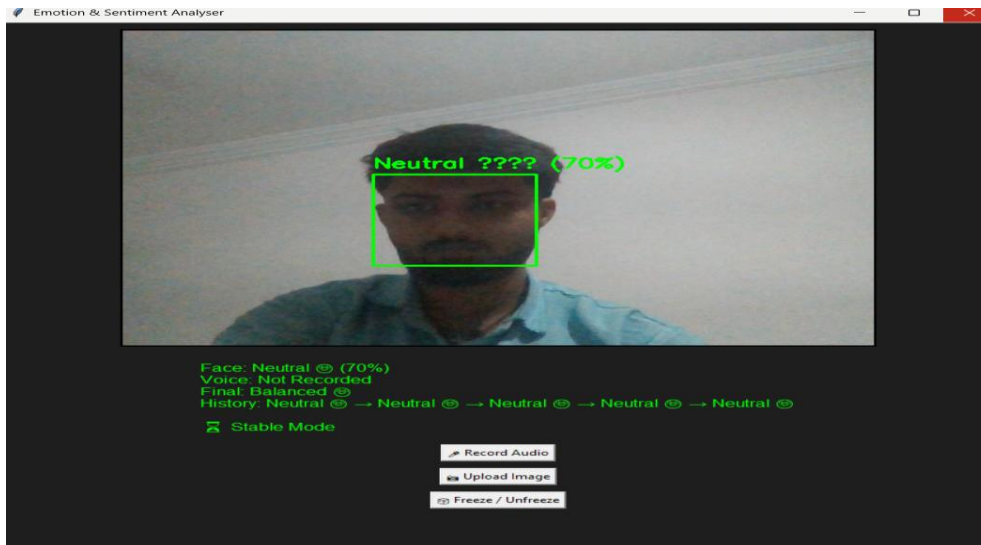
The system successfully performs real-time emotion detection using both facial expressions and voice sentiment. The combination of two inputs provides better

results compared to single-input systems.

However, the accuracy of facial emotion detection is limited because it is based on heuristic methods rather than trained models. Gesture detection also depends on environmental conditions.

Despite these limitations, the system works reliably as a prototype and demonstrates the concept of multimodal emotion analysis effectively.





11. Conclusion

The Emotion & Sentiment Analyzer was developed with the aim of creating a system that can understand human emotions in a more natural way. Instead of relying on a single input, it combines facial expressions and voice to provide a more complete analysis.

The focus during development was on making the system practical and stable rather than overly complex. Features like gesture control, freeze mode, and real-time processing were added to improve usability and interaction.

This project shows that even simple techniques, when combined effectively, can create meaningful and intelligent systems. It also highlights the importance of multimodal approaches in building better human-computer interfaces.

In the future, this system can be improved using deep learning models and advanced tracking methods to increase accuracy and reliability.

12. References

- OpenCV Documentation
- TextBlob Documentation
- SpeechRecognition Library
- Online tutorials and GitHub examples

13. Gratitude

I would like to sincerely thank my teacher for their continuous guidance, support, and valuable feedback throughout the development of this project. Their insights helped shape this work into a more structured and meaningful system. This project is not just about detecting emotions, but about taking a step towards building systems that understand humans better.